Sheet II Problem 11 | Problems T4_PhaseDiagrams II.pdf #11 | Ternary · reading a composition

# Composition of a Ternary Alloy

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 3

*11.- Determine the composition in wt% of the ternary alloy H marked in the diagram below:*

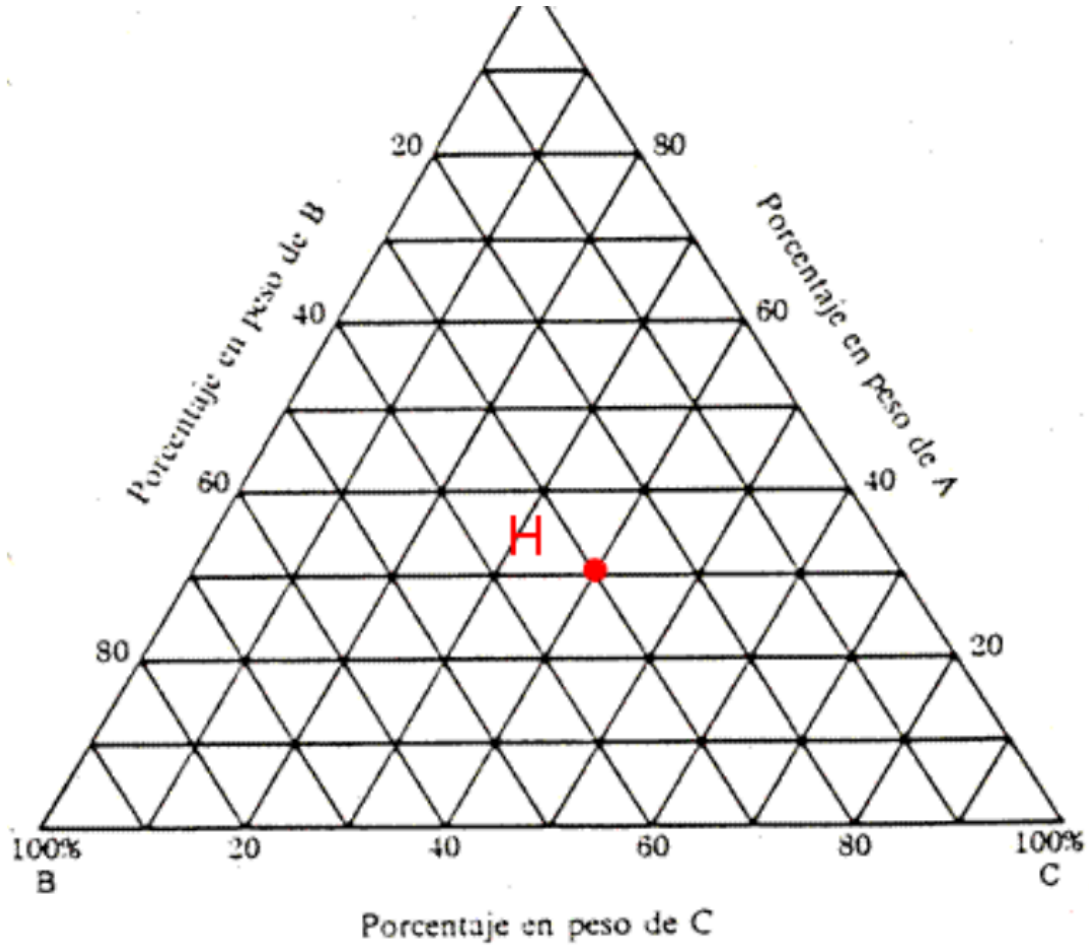



*Figure 41. Ternary diagram with the alloy H (Problems T4_PhaseDiagrams II.pdf, p. 3).*

**Official Solution: 40% C + 33% A + 30%B**

PHASE 1 · HYPOTHESES & PARAMETERS

1. The diagram is a ternary composition triangle in weight percent: corners A (top), B (bottom left) and C (bottom right); the grid lines are every 10 % [Session 10 Slide 5].
2. The axes, as labelled on the sheet: "porcentaje en peso de A" on the right side, "de B" on the left side, "de C" on the bottom.
3. H is drawn on a node of the grid (a reading of the figure).
4. Unknowns: $w_A, w_B, w_C$, with $w_A + w_B + w_C = 100\ \%$.

PHASE 2 · FRAMES & STRATEGY

**Frame:** the triangle itself. Each component is read along the lines parallel to the side opposite its corner: constant A is parallel to B-C, constant B parallel to A-C, constant C parallel to A-B [Session 10 Slides 5, 7]. The three readings are independent, so the condition that they add up to 100 % is a check of the reading, not an extra equation.

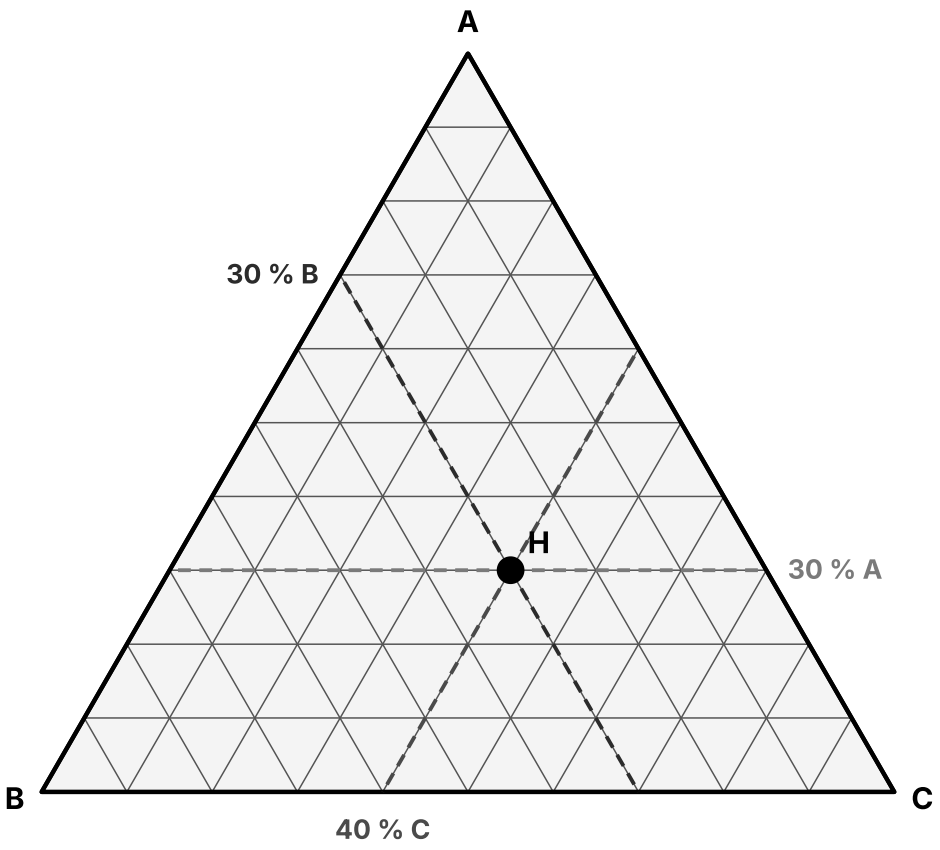



Figure 40. Alloy H of the problem sheet: the lines through H parallel to the sides give 30 % A, 30 % B and 40 % C.

PHASE 3 · STEP-BY-STEP DERIVATION

1. **% A.** The horizontal line through H (parallel to B-C) is the third line above the base B-C: it meets the A axis (right side) at **30 % A**.
2. **% B.** The line through H parallel to the A-C side meets the B axis (left side) at **30 % B**.
3. **% C.** The line through H parallel to the A-B side meets the C axis (bottom) at **40 % C**.

**Composition of H: 30 wt% A, 30 wt% B, 40 wt% C.**

PHASE 4 · CHECKS & INTERPRETATION

- Sum: $30 + 30 + 40 = 100\ \%$.
- Position check: on the 30 % A line, the remaining 70 % is shared between B and C; H is at $40/70 = 0.57$ of the way from the B side to the C side of that line, which agrees with its place on the drawing.
- Limits: a point on a side has 0 % of the opposite component; a corner is 100 % of its component.

**Check against the official solution**

> DISCREPANCY WITH THE OFFICIAL SOLUTION
>
> The key gives "40% C + 33% A + 30% B", which adds up to 103 %, so it cannot be right. H lies on the 30 % A line of the grid (the third line above B-C), and with 30 % B and 40 % C the sum is 100 %. The value 33 % A is a typo for 30 %; the reading 30/30/40 is kept.